

Challenge - Space Missions & Systems

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1 Part for recovering Homework 1

1.1 System dynamics and matrices

The total acceleration $\ddot{\mathbf{r}}$ affecting the spacecraft is the one in Eq.1.

$$\ddot{\mathbf{r}} = \mathbf{a}_{\text{grav}} + \mathbf{a}_{\text{drag}} = -\frac{\mu}{r^3}\mathbf{r} + \frac{1}{2}\rho(z)C_d\frac{A}{m}V^2\hat{\mathbf{V}} \quad (1)$$

In order to estimate the value of the perturbing acceleration due to the atmospheric drag $\mathbf{a}_{\text{drag}}$ an exponential model defined as in Eq. 2 is used.

$$\rho(z) = \rho(z_0) \exp\left(-\frac{z - z_0}{H}\right) \quad (2)$$

1.1.1 Jacobian matrix A

Considering the state vector $\mathbf{X}_{6 \times 1} = [x, y, v_x, v_y, \rho(z_0), C_d]^T$, we get the state transition matrix A in Eq. 4 defined from the dynamical system $\dot{\mathbf{X}} = \mathbf{A}\mathbf{X}$ specified in Eq. 3.

$$\begin{cases} \dot{x} &= v_x \\ \dot{y} &= v_y \\ \dot{v}_x &= -\frac{\mu}{r^3}x - \frac{1}{2}\rho C_d \frac{A_{\text{eff}}}{m} V v_x \\ \dot{v}_y &= -\frac{\mu}{r^3}y - \frac{1}{2}\rho C_d \frac{A_{\text{eff}}}{m} V v_y \\ \dot{\rho}_0 &= 0 \\ \dot{C}_d &= 0 \end{cases} \quad (3)$$

$$A_{6 \times 6} = \frac{\partial \dot{\mathbf{X}}}{\partial \mathbf{X}} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ A_{31} & A_{32} & A_{33} & A_{34} & A_{35} & A_{36} \\ A_{41} & A_{42} & A_{43} & A_{44} & A_{45} & A_{46} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (4)$$

The values in the third row of A are defined as follows:

$$\begin{aligned}
A_{31} &= -\frac{\mu}{r^3} + \frac{3\mu x^2}{r^5} + \left(\frac{C_d A \rho V}{2m H r} \right) v_x x \\
A_{32} &= \frac{3\mu xy}{r^5} + \left(\frac{C_d A \rho V}{2m H r} \right) v_x y \\
A_{33} &= -C_d \frac{A}{2m} \rho \left(V + \frac{v_x^2}{V} \right) \\
A_{34} &= -C_d \frac{A}{2m} \rho \left(\frac{v_x v_y}{V} \right) \\
A_{35} &= -C_d \frac{A}{2m} \exp \left(-\frac{r - R_E - z_0}{H} \right) V v_x \\
A_{36} &= -\frac{A}{2m} \rho V v_x
\end{aligned}$$

The values in the fourth row of A are defined as follows:

$$\begin{aligned}
A_{41} &= \frac{3\mu xy}{r^5} + \left(\frac{C_d A \rho V}{2m H r} \right) v_y x \\
A_{42} &= -\frac{\mu}{r^3} + \frac{3\mu y^2}{r^5} + \left(\frac{C_d A \rho V}{2m H r} \right) v_y y \\
A_{43} &= -C_d \frac{A}{2m} \rho \left(\frac{v_x v_y}{V} \right) \\
A_{44} &= -C_d \frac{A}{2m} \rho \left(V + \frac{v_y^2}{V} \right) \\
A_{45} &= -C_d \frac{A}{2m} \exp \left(-\frac{r - R_E - z_0}{H} \right) V v_y \\
A_{46} &= -\frac{A}{2m} \rho V v_y
\end{aligned}$$

In the first case since we consider as state vector $\mathbf{X}_{5 \times 1} = [x, y, v_x, v_y, \rho(z_0)]^T$, not considering Cd as a state variable to be estimated in the filter, the matrix A is defined as the principal 5×5 submatrix (the top-left minor) of the extended 6×6 Jacobian matrix, where the aerodynamic coefficient C_d is treated strictly as a known constant parameter evaluated at its nominal a priori value. A assumes the form in Eq. 5.

$$A_{5 \times 5} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ A_{31} & A_{32} & A_{33} & A_{34} & A_{35} \\ A_{41} & A_{42} & A_{43} & A_{44} & A_{45} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (5)$$

1.1.2 Computed observables

In order to evaluate the computed observables from the state vector components we use the relations in Eq. 6.

$$\begin{cases} \rho &= \sqrt{(x - R_E \cos \theta)^2 + (y - R_E \sin \theta)^2} \\ \dot{\rho} &= \frac{(x - R_E \cos \theta)v_x + (y - R_E \sin \theta)v_y}{\rho} \end{cases} \quad (6)$$

where:

- R_E is Earth radius;
- θ is the angular position of the station on Earth surface;
- (x, y) e (v_x, v_y) are the state vector components for position and velocity.

1.1.3 Mapping matrix $\tilde{H}$

The mapping matrix $\tilde{H}_{2 \times 6}$ for the state vector $\mathbf{X}_{6 \times 1} = [x, y, v_x, v_y, \rho(z_0), C_d]^T$ is evaluated from the observables relations as in Eq. 7.

$$\tilde{H}_{2 \times 6} = \left[\frac{\partial \mathbf{G}}{\partial \mathbf{X}} \right]^* = \begin{bmatrix} \tilde{H}_{11} & \tilde{H}_{12} & 0 & 0 & 0 & 0 \\ \tilde{H}_{21} & \tilde{H}_{22} & \tilde{H}_{23} & \tilde{H}_{24} & 0 & 0 \end{bmatrix} \quad (7)$$

The relations for the matrix elements are the following:

$$\begin{aligned} \tilde{H}_{11} &= \frac{x - R_E \cos \theta}{\rho} \\ \tilde{H}_{12} &= \frac{y - R_E \sin \theta}{\rho} \\ \tilde{H}_{21} &= \frac{v_x - \dot{\rho} \cdot \tilde{H}_{11}}{\rho} \\ \tilde{H}_{22} &= \frac{v_y - \dot{\rho} \cdot \tilde{H}_{12}}{\rho} \\ \tilde{H}_{23} &= \frac{x - R_E \cos \theta}{\rho} = \tilde{H}_{11} \\ \tilde{H}_{24} &= \frac{y - R_E \sin \theta}{\rho} = \tilde{H}_{12} \end{aligned}$$

In the case of the state vector $\mathbf{X}_{5 \times 1} = [x, y, v_x, v_y, \rho(z_0)]^T$ we get a mapping matrix $\tilde{H}_{2 \times 5}$ which is a submatrix of the extended mapping matrix $\tilde{H}_{2 \times 6}$.

$$\tilde{H}_{2 \times 5} = \left[\frac{\partial \mathbf{G}}{\partial \mathbf{X}} \right]^* = \begin{bmatrix} \tilde{H}_{11} & \tilde{H}_{12} & 0 & 0 & 0 \\ \tilde{H}_{21} & \tilde{H}_{22} & \tilde{H}_{23} & \tilde{H}_{24} & 0 \end{bmatrix} \quad (8)$$

1.2 Results

The state estimation is performed using a batch weighted least-squares (WLS) algorithm with a priori information.

1 - Using both range and range-rate observables provided by the ground station (found in the file “observable.txt”), obtain an estimate of the five components of the state vector and the associated uncertainties at the reference epoch.

Running the code with the putative accuracies of 1 cm and 1 mm/s respectively for range and range rate we get that the filter converges after 3 iterations, but we get values for the SOS which are not acceptable ($SOS_{range} = 9.359e + 04$ and $SOS_{rate} = 8.779e + 04$). For this values we get the estimate of the five state vector components after 5 iterations at the reference epoch $t = 0$ in Table 1 and the residuals plot after 5 iterations in Fig. 1.

Table 1: Final state estimate after 5 iterations at reference epoch ($t = 0$) for range and range rate accuracies of 1 cm and 1 mm/s.

Parameter	Estimated Value	1- σ Uncertainty
x_0 [km]	4.722059×10^3	1.463839×10^{-5}
y_0 [km]	4.722059×10^3	1.482193×10^{-5}
v_{x0} [km/s]	-6.228771	1.375024×10^{-8}
v_{y0} [km/s]	6.228771	1.391135×10^{-8}
$[\rho(z_0)]_0$ [kg/km ³]	1.708272×10^{-2}	3.514022×10^{-7}

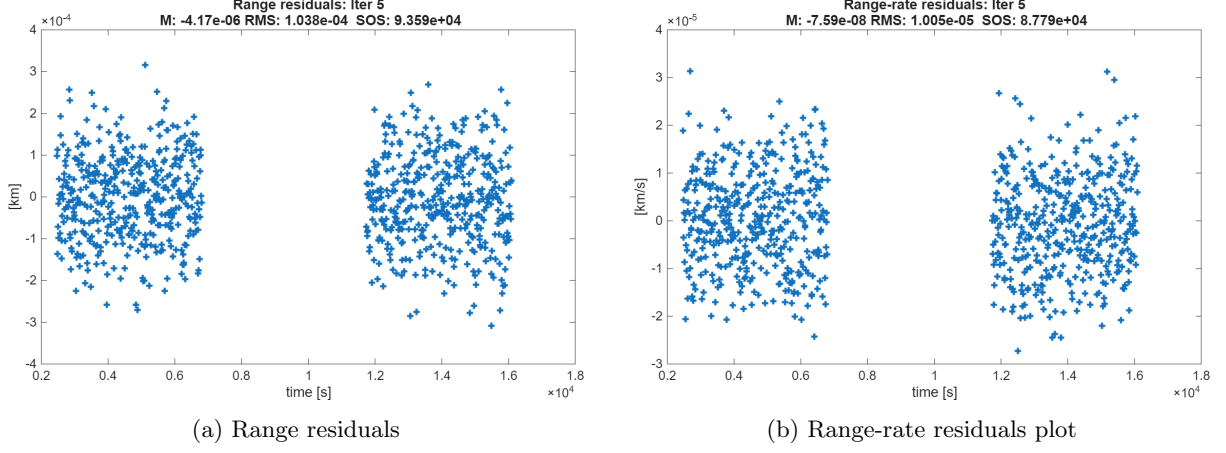


Figure 1: Residuals plot for the observables after 5 iterations for range and range rate accuracies of 1 cm and 1 mm/s.

As we can see the actual accuracies of the range and range-rate observables are $1.043\text{e-}4$ and $1.005\text{e-}5$ respectively (one order of magnitude worse than the putative values). By running the estimation algorithm with these accuracies we get the state estimate after 5 iterations at the reference epoch in Table 2.

Table 2: Final state estimate after 5 iterations at reference epoch ($t = 0$) for range and range rate accuracies of 1.043×10^{-4} km and 1.005×10^{-5} km/s.

Parameter	Estimated Value	1- σ Uncertainty
x_0 [km]	4.722060×10^3	1.526388×10^{-4}
y_0 [km]	4.722059×10^3	1.545526×10^{-4}
v_{x0} [km/s]	-6.228771	1.433777×10^{-7}
v_{y0} [km/s]	6.228771	1.450578×10^{-7}
$[\rho(z_0)]_0$ [kg/km ³]	1.708184×10^{-2}	3.665109×10^{-6}

Here are the residuals plot at iteration 5 with the updated accuracies. As we can see we get a good value for the SOS , indicating a good estimation of the filter.

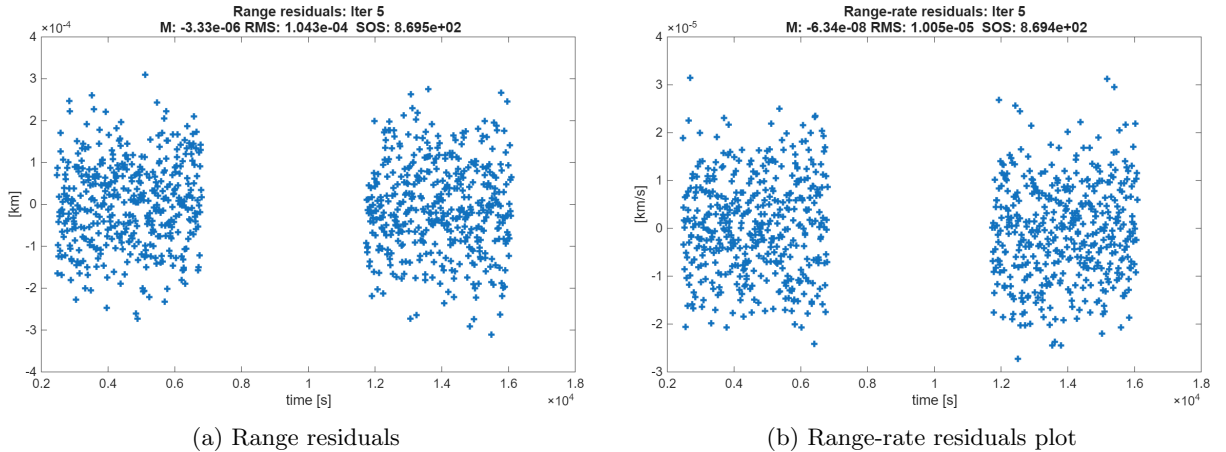


Figure 2: Residuals plot for the observables after 5 iterations for range and range rate accuracies of 1.043×10^{-4} km and 1.005×10^{-5} km/s.

2 - Which of the two types of observables is most valuable for the determination of the probe state? Justify your answer.

In order to find out which observable is the most valuable we run the algorithm twice: the first time using only the range observed values and the second time using only range-rate observed values. In this way the values in Table 3 are obtained.

Table 3: Comparison of Uncertainty Analysis

(a) Range observations only		(b) Range-rate observations only	
Parameter	1- σ Uncert.	Parameter	1- σ Uncert.
x_0 [km]	1.527×10^{-4}	x_0 [km]	4.841×10^{-3}
y_0 [km]	1.546×10^{-4}	y_0 [km]	5.263×10^{-3}
v_{x0} [km/s]	1.434×10^{-7}	v_{x0} [km/s]	4.741×10^{-6}
v_{y0} [km/s]	1.451×10^{-7}	v_{y0} [km/s]	5.091×10^{-6}
$[\rho(z_0)]_0$ [kg/km ³]	3.665×10^{-6}	$[\rho(z_0)]_0$ [kg/km ³]	4.201×10^{-4}

By comparing the two we can see that the uncertainties obtained using only range observations in Table 3a are far more accurate than the ones obtained using only range-rate observations in Table 3b.

We can conclude that range observations are more valuable than range-rate observations.

3 - Considering a new estimation procedure, including also the drag coefficient C_d in the estimated vector with an “a priori” value of 2.014 and an a priori accuracy of 0.01, generate a new orbit determination solution, commenting the final results.

Running the code adding the drag coefficient C_d as a state component and using the updated accuracies for the range and range-rate observables, the filter converges in 3 iterations as the previous case. The final state estimate for the state vector components after 5 iterations are reported in Table 4 and the corresponding residual plots are reported in Fig. 3.

Table 4: Final state estimate at reference epoch ($t = 0$) - Extended State Vector

Parameter	Estimated Value	1- σ Uncertainty
x_0 [km]	4.722060×10^3	1.526388×10^{-4}
y_0 [km]	4.722059×10^3	1.545526×10^{-4}
v_{x0} [km/s]	-6.228771	1.433777×10^{-7}
v_{y0} [km/s]	6.228771	1.450578×10^{-7}
$[\rho(z_0)]_0$ [kg/km ³]	1.696310×10^{-2}	8.430451×10^{-5}
C_{d0}	2.014000	9.999996×10^{-3}

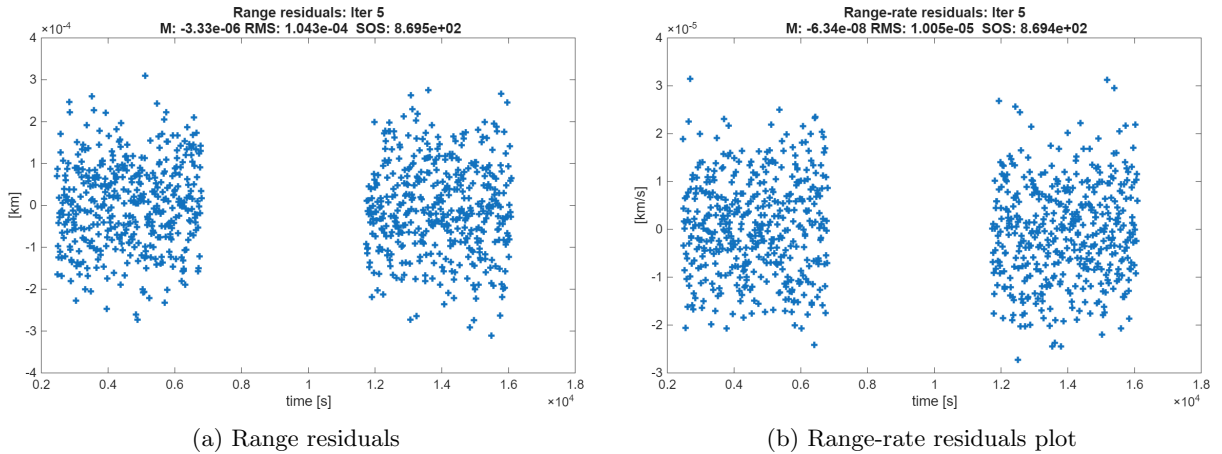


Figure 3: Residuals plot for the observables after 5 iterations for range and range rate accuracies of 1.043×10^{-4} km and 1.005×10^{-5} km/s.

The notable difference between the two cases is the $1 - \sigma$ uncertainty for $[\rho(z_0)]_0$. In the first case, when $\dim(\mathbf{X}) = 5$, the $1 - \sigma$ uncertainty is 3.665109×10^{-6} , while in the second case, when $\dim(\mathbf{X}) = 6$, the $1 - \sigma$ uncertainty is 8.430451×10^{-5} , one order of magnitude larger.

2 Part C: Extended Kalman Filter navigation

In the nominal phase the spacecraft estimates its own state with an Extended Kalman Filter (EKF) processing the pseudo-range and pseudo-range rate measured from the Galileo E11 satellite. The dynamical model is unchanged with respect to Part 1 (Earth monopole gravity and atmospheric drag with $C_D = 2$). All quantities are expressed in km, km/s, s and kg/km³, with $c = 299792.458$ km/s.

2.1 Augmented state, dynamics and state transition matrix

The state is augmented with the clock bias δt and the clock drift $\dot{\delta t}$ of the spacecraft receiver:

$$\mathbf{X}_{7 \times 1} = [x, y, v_x, v_y, \rho(z_0), \delta t, \dot{\delta t}]^T. \quad (9)$$

The clock is modelled as $\delta t(t) = \delta t_0 + \dot{\delta t}(t - t_{0,GNSS})$ with $t_{0,GNSS} = 16100$ s, hence $\dot{\delta t} = \dot{\delta t}$ and $\ddot{\delta t} = 0$. The Jacobian matrix of the dynamics is the 5×5 matrix of Part 1 extended with the clock block:

$$A_{7 \times 7} = \begin{bmatrix} A_{5 \times 5} & 0_{5 \times 2} \\ 0_{2 \times 5} & \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \end{bmatrix}. \quad (10)$$

At each filter step the state and the state transition matrix are integrated together from t_{k-1} to t_k :

$$\dot{\Phi}(t, t_{k-1}) = A \Phi(t, t_{k-1}), \quad \Phi(t_{k-1}, t_{k-1}) = I_{7 \times 7}, \quad (11)$$

using `ode113` with $\text{RelTol} = \text{AbsTol} = 10^{-13}$.

2.2 GNSS ephemeris, computed observables and mapping matrix

The E11 satellite moves on a circular orbit in the x - y plane ($a = 29599.8$ km, $\Omega = 77.632^\circ$, $\omega = 0^\circ$, $e = 0$, $\nu_0 = 15.153^\circ$ at the reference epoch $t = 0$). Its position and velocity at time t are

$$u(t) = \Omega + \nu_0 + nt, \quad n = \sqrt{\mu/a^3}, \quad \mathbf{r}_G = a \begin{bmatrix} \cos u \\ \sin u \end{bmatrix}, \quad \mathbf{v}_G = \sqrt{\frac{\mu}{a}} \begin{bmatrix} -\sin u \\ \cos u \end{bmatrix}. \quad (12)$$

Defining $\Delta x = x - x_G$, $\Delta y = y - y_G$, $\Delta v_x = v_x - v_{x,G}$, $\Delta v_y = v_y - v_{y,G}$ and $\rho = \sqrt{\Delta x^2 + \Delta y^2}$, the computed observables are

$$\tilde{\rho} = \rho + c \delta t, \quad \dot{\tilde{\rho}} = \frac{\Delta x \Delta v_x + \Delta y \Delta v_y}{\rho + c \delta t} + c \dot{\delta t}. \quad (13)$$

With $D = \rho + c \delta t$ and $N = \Delta x \Delta v_x + \Delta y \Delta v_y$, the non-zero elements of the mapping matrix $\tilde{H}_{2 \times 7}$ are

$$\tilde{H}_{11} = \frac{\Delta x}{\rho}, \quad \tilde{H}_{12} = \frac{\Delta y}{\rho}, \quad \tilde{H}_{16} = c, \quad (14)$$

$$\tilde{H}_{21} = \frac{\Delta v_x}{D} - \frac{N \Delta x}{\rho D^2}, \quad \tilde{H}_{22} = \frac{\Delta v_y}{D} - \frac{N \Delta y}{\rho D^2}, \quad \tilde{H}_{23} = \frac{\Delta x}{D}, \quad (15)$$

$$\tilde{H}_{24} = \frac{\Delta y}{D}, \quad \tilde{H}_{26} = -\frac{c N}{D^2}, \quad \tilde{H}_{27} = c. \quad (16)$$

2.3 EKF algorithm

At each measurement epoch t_k the filter performs the following steps:

$$\mathbf{X}_k^- = \varphi(t_k; t_{k-1}, \mathbf{X}_{k-1}^+), \quad P_k^- = \Phi_k P_{k-1}^+ \Phi_k^T, \quad (17)$$

$$\mathbf{y}_k = \mathbf{Y}_k - G(\mathbf{X}_k^-, t_k), \quad S_k = \tilde{H}_k P_k^- \tilde{H}_k^T + R, \quad (18)$$

$$K_k = P_k^- \tilde{H}_k^T S_k^{-1}, \quad \mathbf{X}_k^+ = \mathbf{X}_k^- + K_k \mathbf{y}_k, \quad (19)$$

$$P_k^+ = (I - K_k \tilde{H}_k) P_k^-, \quad (20)$$

where φ is the integrated dynamics, $\mathbf{Y}_k$ the measured pseudo-range and pseudo-range rate, G the computed observables and

$$R = \text{diag}(\sigma_\rho^2, \sigma_{\dot{\rho}}^2), \quad \sigma_\rho = 10 \text{ cm} = 10^{-4} \text{ km}, \quad \sigma_{\dot{\rho}} = 1 \text{ cm/s} = 10^{-5} \text{ km/s}. \quad (21)$$

No process noise is added, since the dynamical model is not changed. The filter starts at $t_{0,GNSS} = 16100$ s, while the first measurement is at 21070 s: the first step is therefore a pure prediction of 4970 s. From there on, one update is performed every 10 s up to 27220 s (616 epochs). The residuals are defined as observed minus computed, before (pre-fit) and after (post-fit) the update.

2.4 A priori information

The a priori state and covariance of the five dynamical components are the final least-squares estimate of Part 1 (Table 2, accuracies 1.043×10^{-4} km and 1.005×10^{-5} km/s), propagated from $t = 0$ to $t_{0,GNSS}$ together with its full covariance matrix, including the correlations:

$$\mathbf{X}(t_{0,GNSS}) = \varphi(t_{0,GNSS}; 0, \mathbf{X}_0), \quad P(t_{0,GNSS}) = \Phi(t_{0,GNSS}, 0) P_0 \Phi(t_{0,GNSS}, 0)^T. \quad (22)$$

The clock parameters, already referred to $t_{0,GNSS}$, are appended and assumed uncorrelated with the orbit:

$$P_0^{EKF} = \text{blkdiag}(P_{5 \times 5}(t_{0,GNSS}), \sigma_{\delta t}^2, \sigma_{\delta \dot{t}}^2). \quad (23)$$

The resulting a priori values are reported in Table 5.

Table 5: A priori state and 1σ uncertainty at $t_{0,GNSS} = 16100$ s.

Parameter	A priori value	1σ uncertainty
x [km]	1.563299×10^3	7.500×10^{-5}
y [km]	-1.045030×10^4	6.220×10^{-5}
v_x [km/s]	5.264051	5.053×10^{-9}
v_y [km/s]	2.439892	6.457×10^{-8}
$\rho(z_0)$ [kg/km ³]	1.708184×10^{-2}	3.665×10^{-6}
δt [s]	2×10^{-6}	1
$\delta \dot{t}$ [s/s]	2.1×10^{-8}	1×10^{-3}

2.5 C1: EKF estimate and residuals

Figure 4 shows the estimated state components as a function of time. The straight segment between 16100 s and 21070 s is only the line joining the a priori value with the first estimate, since no measurement is available in that interval: it is not the real trajectory. The orbit is highly elliptical and the spacecraft altitude varies from about 4300 km at the first measurement, up to about 6000 km, down to about 750 km at the end of the arc.

The clock bias grows linearly up to about 1.12×10^{-4} s. The clock drift moves from the a priori value of 2.1×10^{-8} s/s to 1.0×10^{-8} s/s already at the first update, and then stays constant. The density $\rho(z_0)$ changes only by about 10^{-6} kg/km³ (less than its a priori 1σ of 3.7×10^{-6} kg/km³): at these altitudes the drag is negligible, so the density is practically not observable from the GNSS data and its small variations come from the correlations with the other components.

Figure 5 shows the pre-fit and post-fit residuals of the two observables, together with the $\pm 1\sigma$ bands of the measurements. The first pre-fit residual (-16.7 km on the range and -5.8 m/s on the range rate) is out of scale and is due to the error of the a priori clock drift, accumulated over the 4970 s of pure prediction: $(2.1 \times 10^{-8} - 1.0 \times 10^{-8}) \cdot 4970 \cdot c \approx 16.4$ km, plus about 0.3 km from the bias error. After the first update the residuals are at the level of the measurement noise: the post-fit RMS is 3.0 cm for the range and 5.9 mm/s for the range rate, both below the putative accuracies of 10 cm and 1 cm/s. The range-rate residuals have a small positive mean of about 5 mm/s. Pre-fit and post-fit residuals practically coincide after the first epochs, because the predicted state is already very accurate and, without process noise, the measurements have a small weight in the update.

The final EKF estimate at $t = 27220$ s is reported in Table 6.

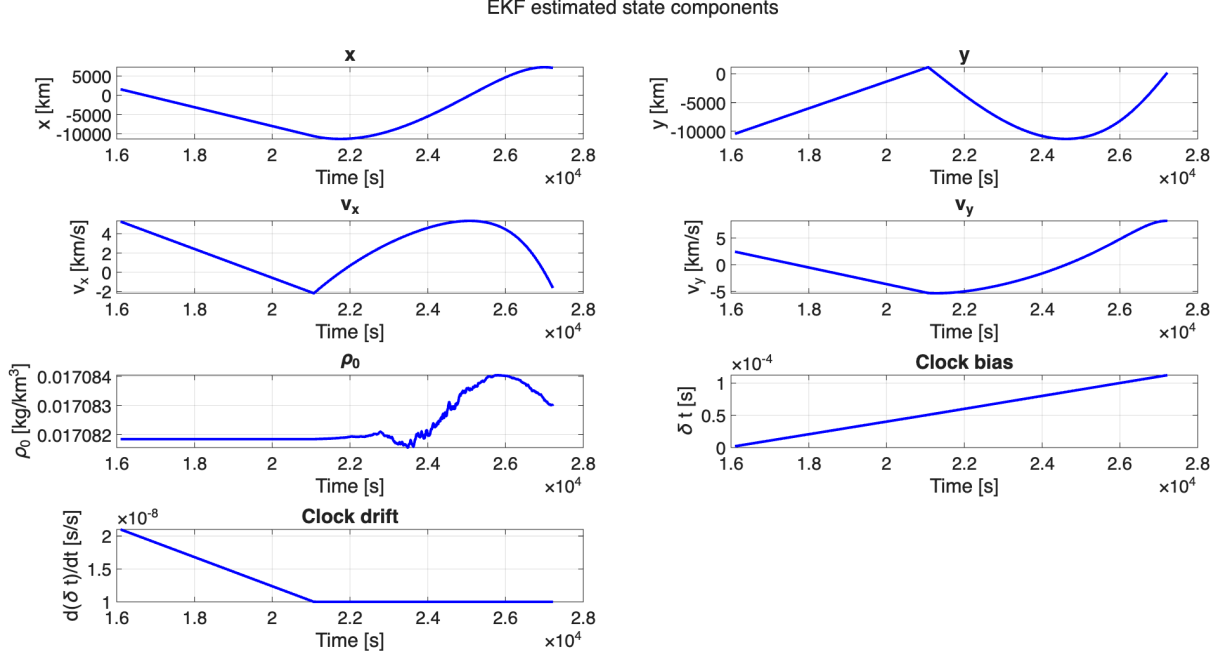


Figure 4: EKF estimated state components as a function of time.

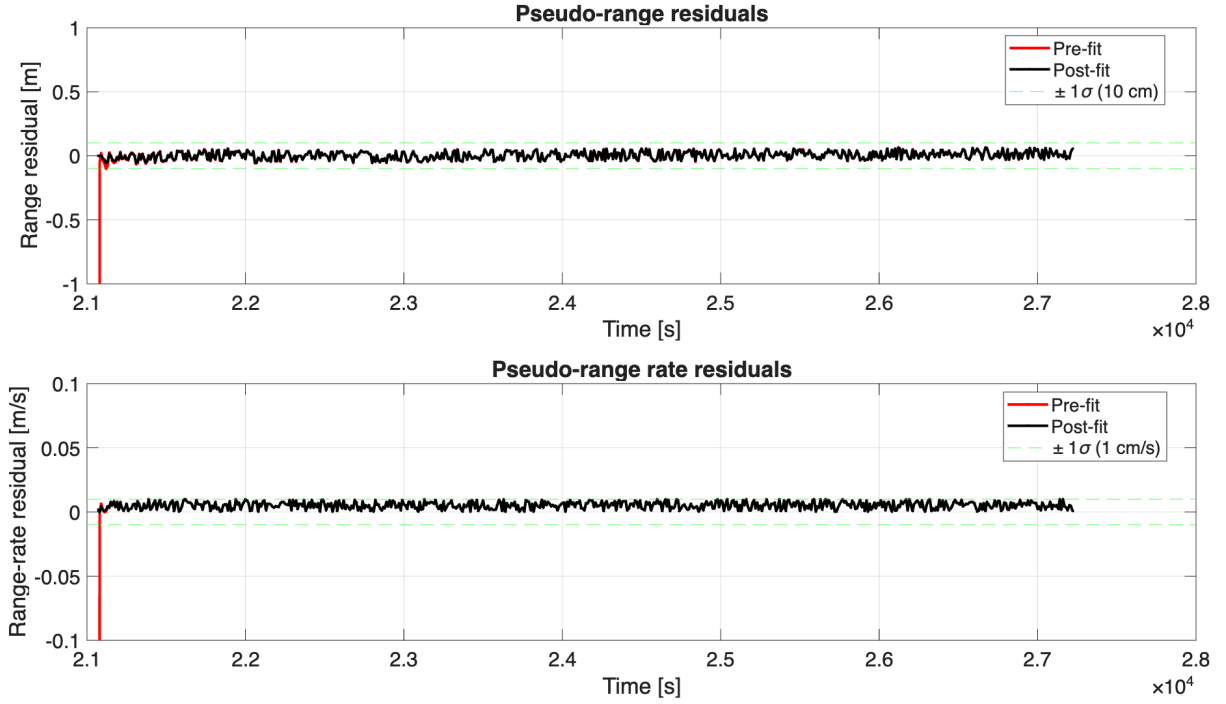


Figure 5: Pre-fit and post-fit residuals of pseudo-range (top) and pseudo-range rate (bottom). The dashed lines are the $\pm 1\sigma$ measurement accuracies.

Table 6: EKF estimate and 1σ uncertainty at the last epoch, $t = 27220$ s.

Parameter	Estimated value	1σ uncertainty
x [km]	7.123848×10^3	1.635×10^{-5}
y [km]	2.046357×10^2	4.504×10^{-5}
v_x [km/s]	-1.631953	4.084×10^{-8}
v_y [km/s]	8.210617	8.210×10^{-9}
$\rho(z_0)$ [kg/km ³]	1.708298×10^{-2}	2.639×10^{-6}
δt [s]	1.122003×10^{-4}	1.322×10^{-10}
$\delta \dot{t}$ [s/s]	1.000002×10^{-8}	4.385×10^{-14}

2.6 C2: Linear regression of the clock bias

The clock bias estimated by the EKF at the 616 measurement epochs is fitted with the linear model

$$\delta t(t) = \delta t_0 + \delta \dot{t} (t - t_{0,GNSS}), \quad (24)$$

where the time is referred to $t_{0,GNSS}$, so that the intercept is directly δt_0 . The least-squares fit gives the values in Table 7, and Figure 6 shows the data, the fitted line and the extrapolated value at $t_{0,GNSS}$.

Table 7: Linear regression of the clock bias (formal 1σ of the least-squares fit).

Parameter	Value	Formal 1σ
δt_0 at $t_{0,GNSS}$ [s]	9.992×10^{-7}	1.4×10^{-11}
$c \delta t_0$ [km]	0.2995	—
$\delta \dot{t}$ (slope) [s/s]	1.0000×10^{-8}	1.7×10^{-15}
Residual RMS of the fit [s]	7.6×10^{-11}	—

The estimated bias at $t_{0,GNSS}$ is $\delta t_0 \simeq 1.0 \times 10^{-6}$ s, i.e. about half of the a priori value (2×10^{-6} s), and the slope 1.0×10^{-8} s/s is consistent with the final drift estimated by the EKF (1.000002×10^{-8} s/s). The formal uncertainties of the fit are optimistic, because the EKF estimates at consecutive epochs are correlated and not independent.

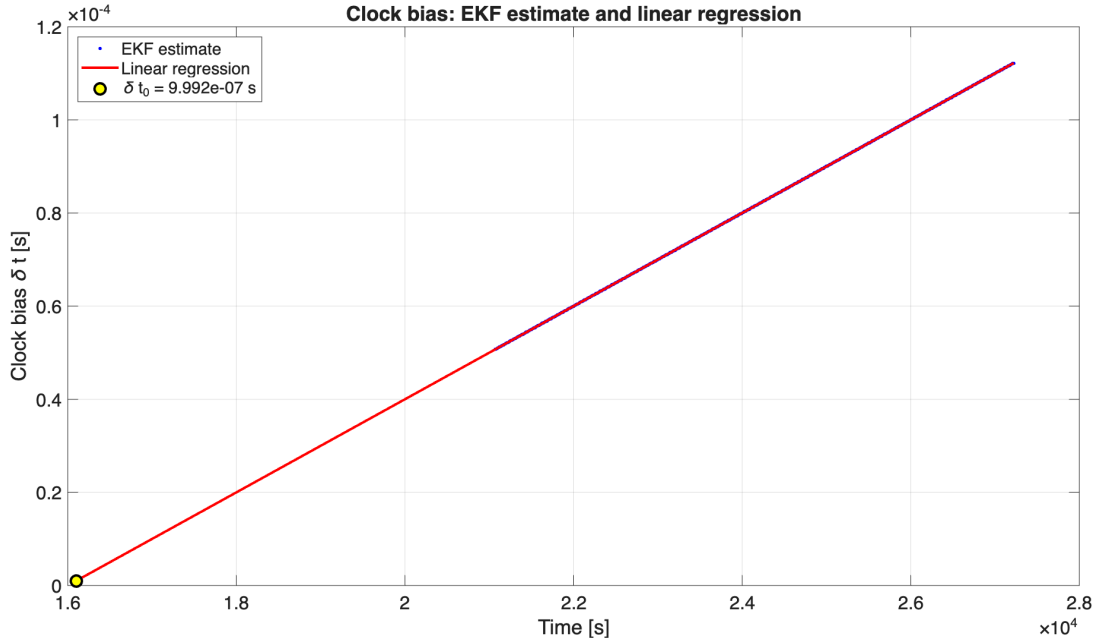


Figure 6: Clock bias estimated by the EKF and linear regression. The yellow marker is the extrapolated value δt_0 at $t_{0,GNSS}$.

2.7 C3: Trace of the covariance matrix

Figure 7 shows the trace of the covariance matrix as a function of time. Between $t_{0,GNSS}$ and the first measurement (blue line) the trace is obtained by propagating the covariance only ($P = \Phi P_0 \Phi^T$, computed on a grid of fictitious epochs for plotting purposes). It increases from 1.0 to about 25.7, dominated by the clock bias variance, which grows as $\sigma_{\delta t}^2(t) = \sigma_{\delta t_0}^2 + \sigma_{\delta \dot{t}}^2(t - t_{0,GNSS})^2$ because of the large a priori uncertainty on the drift. At the first measurement (21070 s) the trace drops to about 1.1×10^{-8} , more than nine orders of magnitude, since the range ($\sigma_\rho = 10$ cm) determines the clock bias immediately. After that the trace slowly oscillates between about 2×10^{-9} and 10^{-8} , with a final value of about 2.3×10^{-9} . Since the trace sums quantities with different units (km^2 , km^2/s^2 , s^2 , ...), the dominant term is the one with the largest numerical value: after the first update it is the position variance.

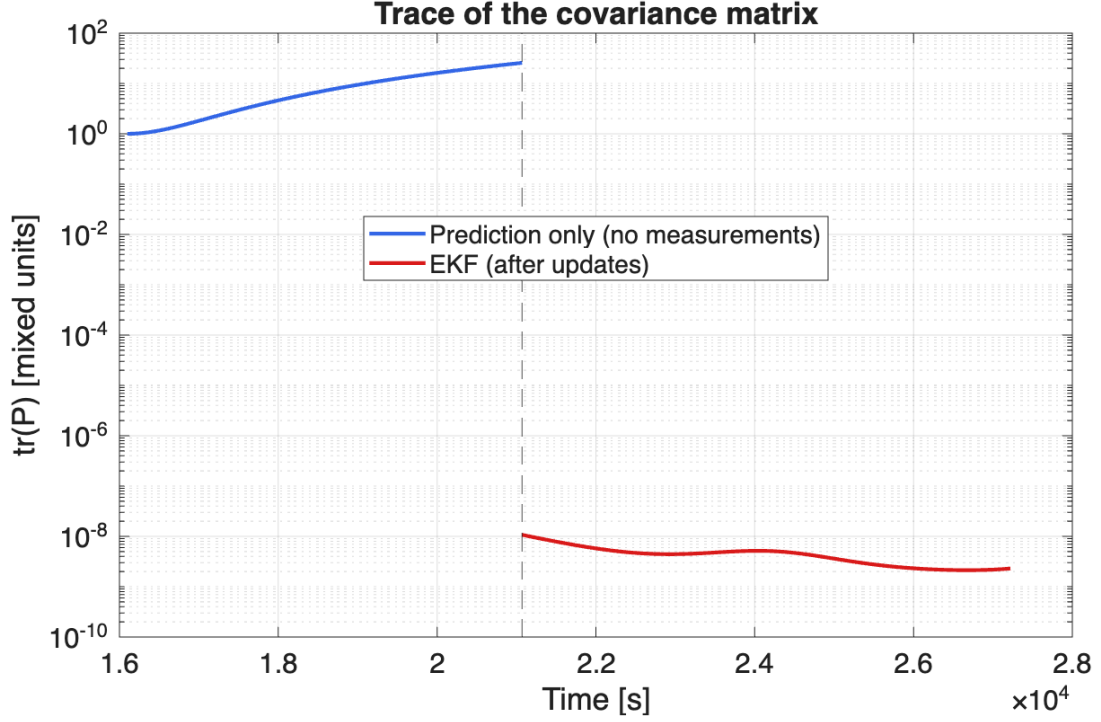


Figure 7: Trace of the covariance matrix: prediction only before the first measurement (blue) and EKF after the updates (red).